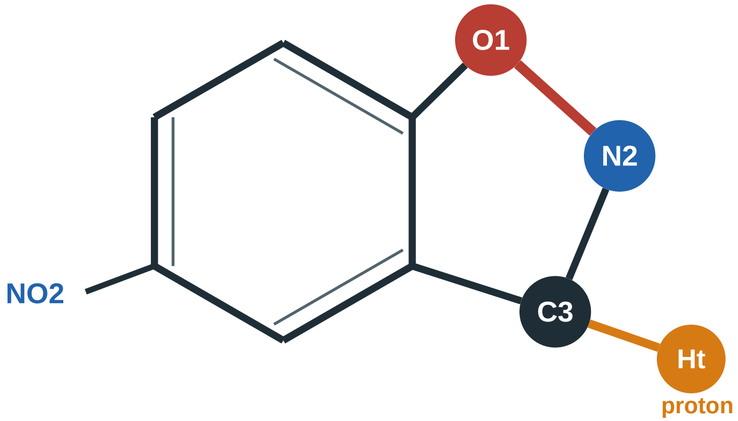
Kemp elimination: proton transfer opens the ring

Asp162 accepts a substrate proton as the N-O bond breaks and the product forms.

REACTANT STATE

5-nitrobenzisoxazole



Asp162
carboxylate

OD1 accepts Ht

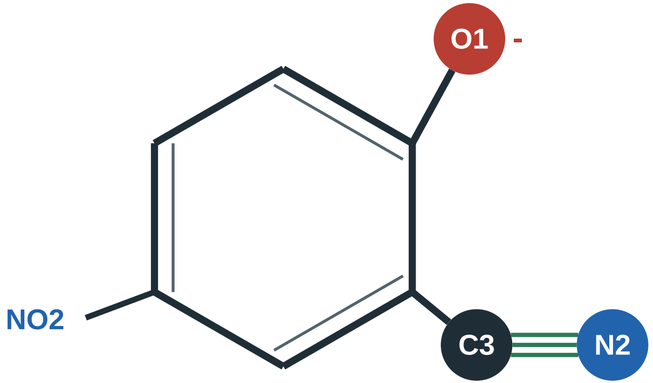
ONE STEP

concerted
bond changes

C3-Ht -> OD1-Ht
N2-O1 breaks
C3-N2 becomes nitrile

PRODUCT STATE

2-cyano-4-nitrophenoxide



Asp162
now protonated

OD1 - Ht

Same atoms; different bonds and proton placement.

Schematic only, not to scale. The paper describes a concerted reaction path; no separate stable intermediate is shown.